

PEDRO BARBOSA

Software Engineer — Python, JavaScript, Go · AI-assisted development

Rio de Janeiro, Brazil (UTC-3) · Remote · barbosamaverickv8@gmail.com · github.com/Amadeus-22 · amadeus-22.github.io/portfolio
Portuguese (native) · English (fluent) · German (basic)

SUMMARY

Engineer who ships production software with a high bar for correctness. Most of my code now starts with coding agents: I run them, review their output, redirect them, and spend my own time on architecture and on deciding what is actually true. I go deep on hard failures — the ones where the reported symptom has three unrelated causes — and I leave automated checks behind so they cannot return silently. Python and JavaScript daily; Go for my own tooling.

EXPERIENCE

Full Stack Developer — TIIX (remote)

April 2026 – present

- One of two developers on a healthcare platform used daily by patients, clinics and professionals.
- Own features end to end: relational schema and versioned migrations, REST endpoints, interface, and the automated tests around them.
- Debug production failures where the reported symptom is not the cause. One example: "the app feels frozen on mobile" turned out to be five independent causes — a hardware check that rejected valid devices, per-frame layout reads, full-screen shader cost, backdrop blur, and the operating system's reduce-motion setting freezing every transition.
- Wrote 23 of the repository's 50 database migrations and the largest share of the automated test suite.
- Work with coding agents daily: I direct them, review every diff, and keep the architecture decisions with me.

Software Engineer — QuantConnect (freelance, remote)

September 2025 – present

- Python on the LEAN engine: backtesting pipelines over large time-series datasets, research models and risk metrics.
- Correctness-critical work: exact numerical handling, deterministic and reproducible results over years of data.

Python Developer — Freelance via Workana (remote)

January 2025 – March 2026

- Django system for a non-profit: role-based access control, email verification, privacy-compliant handling of personal data, normalised schema designed to stay fast as data grows.
- Data pipelines with pandas and openpyxl over spreadsheets (.xlsx), replacing manual processes and removing a recurring source of human error.

Software Development Intern — TT&T Soluções em Informática (remote)

August 2024 – January 2025

- Django backend: orders, authentication, delivery tracking, payment gateway integration.
- MySQL modelling and documentation, unit tests with pytest, optimisation of slow queries and requests.

SELECTED WORK

iod — open source daemon in Go (sole author). Runs external adapters as subprocesses under a versioned contract: bounded timeouts, process-group termination, filtered environments, and validation of every response before it is trusted. Job queue with backoff and orphan recovery. 5,100 lines of code covered by 5,675 lines of tests, with architectural decisions written down.

QuantStrata — risk and exposure engine in Python using exact rational arithmetic, designed so incomplete evidence can never be reported as zero risk: it refuses to produce a number it cannot support. 70 test cases, containerised pipeline, merged through formal code review.

Akaunting — merged contribution to an open source accounting platform used by companies worldwide.

SKILLS

Languages: Python, JavaScript, TypeScript, Go, PHP, SQL, Bash

AI tooling: daily development with Claude Code and other coding agents — multi-step tasks, reviewing generated diffs, context management, keeping architecture decisions outside the agent

Data and documents: MySQL, PostgreSQL, SQLite, pandas, openpyxl (.xlsx), structured-document processing

Engineering: Clean Code and architecture, REST API design, automated testing (unit and integration), code review, versioned migrations, Docker, Git, CI/CD, Linux

EDUCATION AND CERTIFICATIONS

Computer Engineering (Data and Machine Learning) — Centro Universitário IBMR · 2023 – 2028 (expected) · GPA 8.7/10

Scientific initiation in Economic Mathematics and Machine Learning certificate — IMPA, 2024

HackerRank: Problem Solving (Intermediate), REST API (Intermediate), Python · Coursera: Python for Data Science and AI